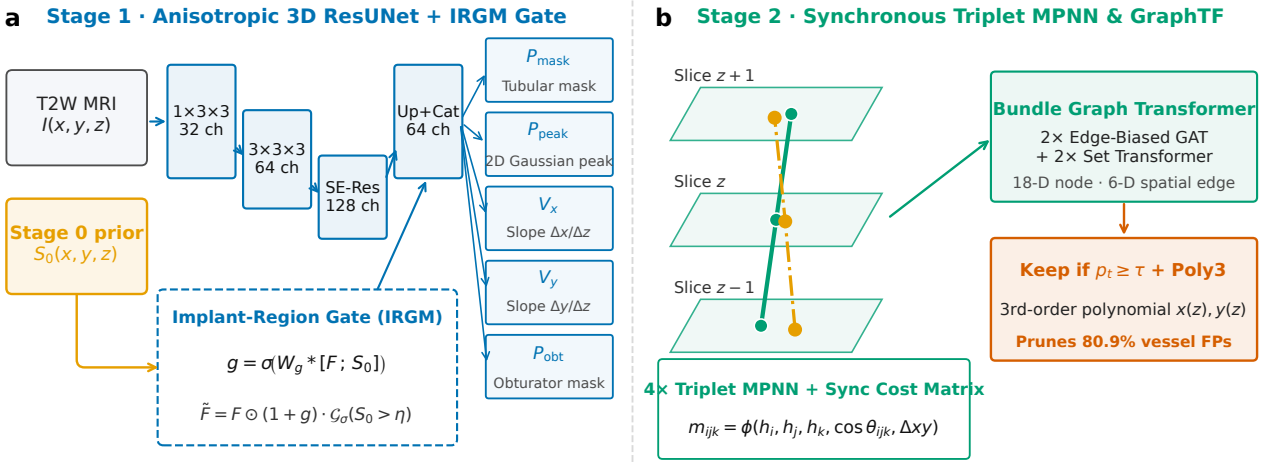
a
Stage 1 · Anisotropic 3D ResUNet + IRGM Gate
T2W MRI
$I(x, y, z)$
1×3×3
32 ch
3×3×3
64 ch
SE-Res
128 ch
Up+Cat
64 ch
P_{mask}
Tubular mask
P_{peak}
2D Gaussian peak
V_x
Slope $\Delta x/\Delta z$
V_y
Slope $\Delta y/\Delta z$
P_{obt}
Obturator mask
Stage 0 prior
$S_0(x, y, z)$
Implant-Region Gate (IRGM)
$g = \sigma(W_g * [F\,;\, S_0])$
$\tilde{F} = F \odot (1 + g) \cdot \mathcal{G}_\sigma(S_0 > \eta)$
b
Stage 2 · Synchronous Triplet MPNN & GraphTF
Slice $z+1$
Slice z
Slice $z-1$
Bundle Graph Transformer
2× Edge-Biased GAT
+ 2× Set Transformer
18-D node · 6-D spatial edge
Keep if $p_t \geq \tau$ + Poly3
3rd-order polynomial $x(z), y(z)$
Prunes 80.9% vessel FPs
4× Triplet MPNN + Sync Cost Matrix
$m_{ijk} = \phi(h_i, h_j, h_k, \cos\theta_{ijk}, \Delta xy)$